

Mock Test — Electricity: Magnetic and Heating Effects

A mixed paper covering the whole chapter. Total: 20 marks. Section A is interactive.

Objective (1 mark each)

Q1 • MCQ

[1]

A current-carrying wire deflects a nearby compass. This shows the _____ effect of current.

- (a) heating (b) magnetic (c) chemical (d) sound

Q2 • MCQ

[1]

An electromagnet becomes stronger if you —

- (a) reduce the current (b) increase the number of turns (c) remove the iron core
(d) use one cell instead of two

Q3 • MCQ

[1]

Nichrome wire is used in heaters because it —

- (a) has high resistance and gives more heat (b) is the best conductor (c) is cheapest
(d) does not get hot

Q4 • MCQ

[1]

A cell generates current due to —

- (a) magnetism (b) chemical reactions inside it (c) heating (d) light

Q5 • MCQ

[1]

In a dry cell, the positive terminal is the —

- (a) zinc container (b) carbon rod (c) electrolyte paste (d) outer wrapper

Q6 · TRUE / FALSE

[1]

Oersted discovered the link between electricity and magnetism.

(a) True (b) False

Q7 · FILL IN THE BLANK

[1]

The liquid used in a Voltaic cell is called the _____.

Q8 · FILL IN THE BLANK

[1]

A current-carrying coil that behaves like a magnet is an _____.

Short answer (2 marks each)

Q9

[2]

State two ways to increase the strength of an electromagnet.

Q10

[2]

Why is copper used for connecting wires and nichrome for heating elements?

Q11

[2]

Match the device to its effect: (i) electric iron (ii) electric bell.

Long / application (3 marks each)

Q12

[3]

Describe an activity to show that a current produces a magnetic field, and state two ways to make the resulting electromagnet stronger.

Q13

[3]

A nichrome wire gets warm and the compass beneath it deflects. Identify both effects shown and explain why each occurs.

[Check yourself](#) → Answer key